1. Bijections

1) $1, 10, 19, \ldots, 2026 \mid +8$

$9, 18, 27, \ldots, 2034 \mid :9$

$1, 2, 3, \ldots, 226 \Rightarrow$ 226 numbers

$100: 1 + 99 \cdot 9 = 891 + 1 = 892$

2) $1^2, 2^2, \ldots, 45^2$ $\quad 1, 4,$

$\downarrow \quad \downarrow \quad \downarrow \Rightarrow$ 45 de nr

$1 \quad 2 \quad 45$

45.
45
225
180
2025

3) 243 $\quad 243 = 3^5$

$100000 \Rightarrow 2^5 - 1 = 31$ de nr

4) $M = \{1, 2, \ldots, 10\}$ $\quad \{1, 2,$

$1, 2, 3, 4, 5, 6, 7 \rightarrow 8, 9, 10$

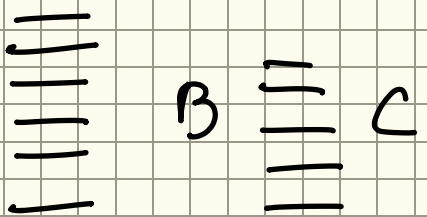
$f(A) = M \setminus A \quad f: A \rightarrow B$

5) $\underline{9}\ \underline{9}\ _\ _\ _\ _$ $\quad \underline{10}\ \underline{9}\ \underline{9}\ \underline{9}\ \underline{9}\ \underline{9}$

$\downarrow \quad \downarrow$ $\quad 9^5 \cdot 90$

6) $\underline{9}\ \underline{10}\ \underline{10}\ \underline{1}\ \underline{1}$ $\quad 10^3 \cdot 9^2$

900



7) $5040 = 2^4 \cdot 3^2 \cdot 5 \cdot 7$

$\underline{5}\ \underline{3}\ \underline{2}\ \underline{2} = 5 \cdot 3 \cdot 2 \cdot 2 = 60$ $\quad \underline{6}\ \underline{5}\ \underline{4}\ \underline{5}$

$6 \cdot 5 \cdot 4 \cdot 5$

$24 \cdot 25 = 600$

8) $A \rightarrow B: 6$

$B \rightarrow C: 5$

25.
24
100
50 6r